Divyansh Shukla

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Education

B.Tech, Computer Science and Engineering (AI and ML) — VIT-AP University, Amaravati
2023 – 2027 | CGPA 9.03 / 10

CBSE — DAV Public School, Sahibabad
XII (2022) – 91% | X (2020) – 95.6%

Experience

Machine Learning Intern — Power Pulse India, Ghaziabad | 2026 – Present
Thermal perception for advanced driver assistance systems

- Trained a YOLOv8n thermal object detector on FLIR ADAS v2 across six road-safety classes, taking mAP50-95 from 0.360 to 0.420 and cutting missed pedestrians by 26%, from 1,352 down to 999.
- Ran a six-experiment sweep over input resolution and architecture. Dropped a P2 detection head that scored below a plain 640px model while costing 1.51x the compute and 4x the detection candidates.
- Audited per-class crops and COCO size buckets to locate the accuracy ceiling. 82% of pedestrian labels sit under 32x32 px, and the weak truck class turned out to be measuring annotation disagreement rather than model error, with identical delivery vans labelled truck, car and bus across the dataset.
- Cut inference compute by 29% at equal accuracy by moving to an NMS-free YOLO26n once the deployment target changed to a 3 TOPS ARM NPU.

Projects

Fashion Search: Conversational Product Discovery | *Python, FastAPI, Gemini, ChromaDB, Tavily*

- Built a FastAPI chat application that answers natural-language shopping queries across 10 Indian e-commerce stores, using Tavily search, Gemini 2.5 Flash extraction and BeautifulSoup scraping.
- Assembles complete outfits instead of returning single items, and lets the user refine them across a multi-turn conversation, with the candidate products browsable in a carousel.
- Wrote the agent loop by hand on the Google Gen AI SDK without a framework. A planner step decides whether to search or ask a clarifying question, conversation history acts as memory, and Gemini function calls are dispatched manually against a declared search tool.

Smart Waste Segregator | *PyTorch, ONNX, onnxruntime, Raspberry Pi, LIDAR*

- Built a Raspberry Pi smart bin that sorts waste into four compartments with a dual-flap servo mechanism. Trained a MobileNetV3-Small to 91.63% validation accuracy, exported it to ONNX at opset 11 with output parity checked against PyTorch, and held trigger-to-actuation under 3 seconds.
- Showed a reported 2.75% test accuracy was a broken harness, not a broken model: six-class YOLO labels read against a four-class head silently dropped 860 of 1,042 images and mislabelled the rest. Recovered the labels from the filenames and re-scored back in line with validation.
- Root-caused the real-world misclassification as distribution mismatch, centred catalogue photos in training against a fixed overhead camera on crumpled trash, and fixed thermal throttling by replacing always-on video decoding with a LIDAR trigger.

RoPE-DiT: Diffusion Transformer with 2D Rotary Embeddings | *PyTorch*

- Wrote a 5M-parameter Diffusion Transformer from scratch: patchifier, timestep and class-label embedders, AdaLN-Zero conditioning and custom multi-head attention. Trained on CIFAR-10 with a rectified flow objective.
- Added a 2D rotary positional embedding that rotates queries and keys by each patch's row and column coordinate, so attention sees relative 2D distance instead of a flattened index.

NagarKranti: Citizen Grievance Platform | *Django, PostgreSQL, PostGIS*

- Built a Django platform where citizens report civic issues and municipal authorities track and resolve them, backed by PostgreSQL with PostGIS for geospatial queries and a custom user model.

POSIX Shell in C++ | *C++, POSIX process APIs*

- Wrote a working Unix shell from scratch: tokenizer, builtin commands, PATH resolution, process spawning and output redirection.

Technical Skills

Languages: Python, C++, CUDA C++, SQL
Core CS: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, OOP, SDLC and Agile
ML and Deep Learning: PyTorch, PyTorch Lightning, Hugging Face Transformers, Ultralytics YOLO, Stable-Baselines3, Gymnasium, MuJoCo, OpenCV, NumPy, pandas
Backend and Data: FastAPI, Flask, Django, REST APIs, PostgreSQL, PostGIS, SQLAlchemy, SQLite
Deployment and Tooling: ONNX, TorchScript, NCNN, edge inference, Docker, Kubernetes (basics), AWS (EC2, S3, Lambda, IAM, RDS, VPC), Google Cloud (BigQuery, Dataflow, Looker Studio), Modal, Weights and Biases, Git, Linux, pytest

Certifications and Achievements

- **Deep Learning for Computer Vision**, NPTEL / IIT Hyderabad. Elite + Topper, top 1% of 463 candidates.
- **Applied Accelerated Artificial Intelligence**, NPTEL / IIT Guwahati and KLE Tech. Elite + Topper, top 2% of 591 candidates.
- **Google Cloud Data Analytics Certificate** (5 courses) and **Introduction to Data Engineering on Google Cloud**, 2026. Deep Learning Specialization, DeepLearning.AI on Coursera. AWS Certified Cloud Practitioner and AWS Academy Cloud Architecting.
- Won the intra-campus round of the Smart India Hackathon at VIT-AP, 2024, in a team of five.